

Electrical Stimulation Mapping during Intracranial EEG Recordings

Policy and Procedure (v1.0)

This policy's purpose is to motivate and explain the electrical stimulation mapping procedure in patients with epilepsy undergoing intracranial EEG monitoring.

Background and rationale

When cortical stimulation is performed in patients with epilepsy following standard protocols, it induces seizures in 34-57% of patients. The presence of stimulation-induced seizures and resecting electrode contacts implicated in the stimulation-induced seizure onset zone are both correlated with good surgical outcomes. Thus, performing cortical stimulation following a standard protocol may add useful ancillary information for surgical planning (Oderiz et al., 2019; Jobst et al., 2020; Trébuchon et al., 2021).

Stimulation-induced seizures are more likely when the patient's last spontaneous seizure was >24 hours prior to stimulation (Oderiz et al., 2019; Jobst et al., 2020). Stimulation-induced seizures are more common with high frequency (50 Hz) stimulation than with low frequency (1 Hz) stimulation. However, atypical seizures are more common with high frequency stimulation (7.8%) than with low frequency stimulation (1.5%) (Oderiz et al., 2019).

The above data support performing both low- and high-frequency stimulation. In particular, the following protocol involves performing low frequency stimulation on most contacts *prior* to weaning anti-seizure medications to maximize the probability of inducing a seizure using an unbiased approach. Then, if no seizures are induced with low frequency stimulation and the epilepsy localization remains uncertain following capture of spontaneous seizures (or if needed for functional mapping), high frequency stimulation should be performed targeting hypothesized seizure onset zone contacts (and relevant eloquent cortex, in the case of functional mapping). This targeted approach to high frequency stimulation minimizes the risk of capturing atypical seizures.

Timing of stimulation

Low frequency stimulation should be performed for all cases *prior* to weaning anti-seizure medications. If the patient is on a long-acting anti-seizure medication at home and this medication was weaned prior to EMU admission, stimulation may be performed in the setting of this wean. Ideally, stimulation should be performed the morning of the second day of intracranial EEG recording according to clinical team availability. This timing is late enough to allow clinicians to capture spontaneous interictal EEG prior to stimulation, and early enough to not interfere with the timing of medication wean. We expect that the low frequency stimulation procedure will take approximately 2 hours (assuming 100 contact pairs are stimulated).

High frequency stimulation should only be performed under the following circumstances:

Stimulation is needed to map eloquent cortex

OR

No seizures were induced with low frequency stimulation **AND** the epilepsy localization remains uncertain after the remainder of the intracranial EEG evaluation.

High frequency stimulation should be performed *after* collecting spontaneous seizures and once the patient has been restarted on home medications. Ideally, it should be performed at least 1 day prior to explant. We expect that high frequency stimulation will take approximately 1 hour (assuming 20-30 contact pairs are stimulated).

Setting Up

Intravenous lorazepam (2 mg) should be available in the room, and the patient should have a working IV. Equipment needed for the procedure is the **Nicolet Cortical Stimulator** box with external power cable and red and black **Daisy Chain** cable.

Plug the power cable for stimulator box into wall outlet and turn the power button on. The Daisy Chain connects to the top of stimulator box via the black and red plugs on one end and the 6 plug female connector plugs into the Natus Quantum box- marked by a lightning bolt. If 2 Quantum break-out boxes are in use, utilize the splitter cable and plug into both "lightning bolt" inputs. *See the Figures at the end of the protocol for a schematic.*

Using the Natus computer in the room with patient study pulled up → Highlight Controls → DSM Stimulation. This opens the stimulator controls and allows to choose 2 electrodes to stimulate between through a drop-down menu.

After choosing the designated 2 electrodes click on the Green check (this opens the pathway for stimulation). When finished click Red "X" to change the designated electrodes. Continue with electrode pairs given by the physician for stimulation.

Stimulation Team

The following tasks need to be performed in stimulation:

- Stimulation task: Setup and breakdown of the stimulator. Adjust stimulation parameters and deliver stimulation
- EEG task: Monitor the EEG for seizures and after discharges, and adjust stimulation contacts
- Clinical task: Monitor the patient for clinical changes and administer IV Ativan if needed
- Recording task: Transcribe Electrode Stimulation sheet (electrodes stimulated, current, frequency, duration, task and effect). Prepare and complete the official Stimulation Report.

At least two team members should always be present in the room to perform all stimulation tasks. The epilepsy attending should always monitor the stimulation session. It is up to the attending's discretion whether to monitor stimulation from the patient room or remotely in the EMU workroom.

Choosing electrodes to stimulate

For **low frequency stimulation**, all adjacent pairs of electrode contacts should be stimulated including white matter contacts, with the following exceptions:

- Contacts outside the brain or adjacent to the meninges
- Contacts with significant artifact or abnormal impedance
- Contacts believed likely to produce a noxious sensation with the planned stimulation parameters

For **high frequency stimulation**, only those electrode contacts hypothesized to be the seizure onset zone or in eloquent cortex (for functional mapping) should be stimulated. The same electrode exclusion criteria apply as with low frequency stimulation.

For both low and high frequency stimulation, the contacts thought most likely to be the seizure onset zone should be stimulated last to avoid inducing a large seizure early in the stimulation session. However, because LFS occurs prior to capturing spontaneous seizures, there is no reliable way of predicting which contacts will yield ictal patterns. Although it is encouraged to collect data from all contact pairs, it is up to

the attending's discretion to continue LFS once a seizure has been induced. Use of the ictal disrupt button is highly encouraged.

Stimulation protocol

Make sure Natus is annotating the stimulation contacts and parameters. The annotation should read "Start stimulation from X to Y ([parameters])"

Low frequency stimulation

- Setup: Probe Biphasic
- Pulse Duration: 500 us
- 1 Hz frequency
- 30 second train duration
- 3 mA current
- Wait until the EEG normalizes post-stimulation before proceeding to the next contact pair

High frequency stimulation and functional mapping

- Setup: Probe Biphasic
- Pulse Duration: 500 us
- 50 Hz frequency
- 5-6 second train duration
- Increase current from 1-4 mA in 1 mA steps in the neocortical contacts, and 1-3 mA in 1 mA steps in the mesial temporal contacts. Per attending discretion, stimulation could instead be performed at 4 mA in the neocortical contacts (3 mA in the mesial temporal contacts) without gradual increase.
- Per attending discretion, higher currents may be used provided they remain under the established safety limits of charge density in order to avoid tissue injury (see safety table below).
- Wait until the EEG normalizes post-stimulation before proceeding to the next current
- Special care should be taken when performing high frequency stimulation in the following circumstances:
 - **Primary motor cortex:** this may produce a strong noxious stimulus
 - **Mesial temporal cortex, Heschl's gyrus, and dysplastic cortex:** seizures and afterdischarges can be induced by high frequency stimulation with lower currents in these regions (Oderiz et al., 2019; Trébouchon and Chauvel, 2016). Seizures induced in these regions with high frequency stimulation should therefore be interpreted with particular caution

Testing functional cortex

When performing high frequency stimulation to hypothesized regions of eloquent cortex, assess for functional changes using the following approach:

For hypothesized receptive or productive language areas:

- Have the patient read a block of text aloud (starting stimulation 2-3 sentences into the text)
- If baseline reading ability is impaired, use counting/reciting familiar lines instead
- If the patient has speech arrest, ask them the reason for the arrest (to rule out other causes such as pain)

For hypothesized visual cortex:

- Place a 16x22 inch sheet of paper 50 cm in front of the patient, with horizontal and vertical lines across the midline and a 3 mm fixation point at the intersection
- Ask the patient to look at the fixation point during stimulus
- Trace the visual response location on the sheet of paper and record the description

For hypothesized sensorimotor cortex:

- Note that complaints of dysesthesias may be subtle motor rather than sensory phenomena (in which case, a higher current will likely elicit a motor response)
- To test for a negative motor response, ask the patient to perform a sustained motor task (such as finger tapping for hypothesized hand motor cortex)

General protocol

Both the patient and the EEG should be monitored throughout each stimulation for seizures, afterdischarges, or other clinical effects. The stimulation current should be documented on Natus. All stimulation parameters and their electroclinical effects should be documented in the Epic report.

End stimulation at a given pair of contacts if it induces a seizure or if it produces an uncomfortable sensation. The “Stop” button can be pressed to abort the 30-second train of low frequency stimulation.

The patient’s seizure button should be pressed for any seizure both for documentation purposes and to alert the remainder of the clinical team to enter the room for additional support. If an electroclinical seizure is prolonged (>30 seconds), the “Ictal Disrupt” button should be pressed. If the seizure continues despite Ictal Disrupt x 2-3 or if the seizure progresses to a convulsion, IV lorazepam should be administered.

If a convulsive seizure is induced, then stimulation should be stopped for the day. Stimulation may proceed at the remaining contacts if a focal aware or focal impaired awareness seizure is induced at the attending’s discretion based on the specific clinical situation. The clinical team should determine whether additional stimulation is needed if the stimulation session is stopped early due to a seizure.

Reports

The epilepsy fellow should document the report in Epic using “.corticalstimreport”.

Billing

To bill for cortical stimulation, use one of the paper bills (similar to those used to bill for EEG reading in the EMU). Ask the lead EMU technician for a paper bill. Use the following CPT codes:

- 95961 (Functional cortical and subcortical mapping by stimulation and/or recording of electrodes on brain surface, first hour)
- 95962 (Functional cortical and subcortical mapping by stimulation and/or recording of electrodes on brain surface, each addtl hour)) – this can be billed once you’ve crossed 31 minutes.

For example, mapping starts 10AM and ends at 12:34PM. The attending will bill:
95961 x 1 (10:00-11:00) + 95962 x 2 (11:00-12:00 & 12:00-12:34)

Billing is based on day of service. If a mapping session is paused and the team returns at a later time to complete the session the same day, add up the time within the day (i.e., cannot bill 95961 twice on the same day). LFS and HFS sessions are billed as separate encounters, so there will be two separate 95961 if both sessions are performed.

Safety parameters to avoid local tissue injury

The primary metric that predicts local tissue injury from electrical cortical stimulation is the charge density:

$$\text{Charge density} \left(\frac{\mu\text{C}}{\text{cm}^2} \right) = \frac{\text{Stimulation intensity (mA)} \times \text{Pulse duration (ms)}}{\text{Electrode surface area (cm}^2\text{)}}$$

Two reported safety limits for charge density are $<30 \mu\text{C}/\text{cm}^2$ (McCreery et al., 1990) and $<55 \mu\text{C}/\text{cm}^2$ (Gordon et al., 1990). The following table shows the calculated charge densities for Penn's current AdTech depth electrodes (as of February 2022):

STIMULATION PARAMETERS

"S" style contact:

Length: 2.41 mm

Diameter: 1.10 mm

Charge Density ($\mu\text{C}/\text{cm}^2$)		Current (mA)												
		0.2	0.4	0.6	0.8	1	1.5	2	3	4	5	7.5	10	15
5	0.1	0.25	0.49	0.74	0.99	1.23	1.85	2.47	3.70	4.94	6.17	9.25	12.34	18.51
	0.2	0.49	0.99	1.48	1.97	2.47	3.70	4.94	7.40	9.87	12.34	18.51	24.68	
	0.3	0.74	1.48	2.22	2.96	3.70	5.55	7.40	11.10	14.81	18.51	27.76		
	0.4	0.99	1.97	2.96	3.95	4.94	7.40	9.87	14.81	19.74	24.68			
	0.5	1.23	2.47	3.70	4.94	6.17	9.25	12.34	18.51	24.68				
	0.75	1.85	3.70	5.55	7.40	9.25	13.88	18.51	27.76					
	1	2.47	4.94	7.40	9.87	12.34	18.51	24.68						
	1.25	3.08	6.17	9.25	12.34	15.42	23.13							
	1.5	3.70	7.40	11.10	14.81	18.51	27.76							
	1.75	4.32	8.64	12.95	17.27	21.59								
	2	4.94	9.87	14.81	19.74	24.68								
	2.5	6.17	12.34	18.51	24.68									
	3	7.40	14.81	22.21	29.61									

$$\text{Charge Density } \frac{\mu\text{C}}{\text{cm}^2} = \frac{\text{Stimulation Intensity (mA)} * \text{Pulse Duration (ms)}}{\text{Electrode Surface Area (cm}^2\text{)}}$$

The electrode surface area is 0.081 cm^2 . The maximum charge density limit for safe tissue stimulation is $30 \mu\text{C}/\text{cm}^2$.

References

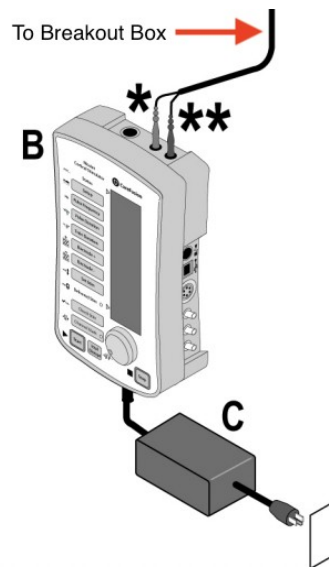
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- Trébuchon, Agnès, and Patrick Chauvel. "Electrical stimulation for seizure induction and functional mapping in stereoelectroencephalography." *Journal of Clinical Neurophysiology* 33.6 (2016): 511-521.

Photos of stimulator setup

Front view of the Nicolet stimulator:



Diagram of stimulator showing all connections:



Power button and connection to power cable:

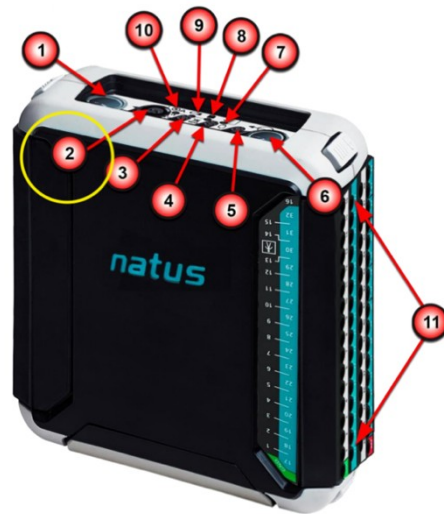


(plugged)

Top of box with red and black outputs to breakout



Natus breakout box (usually in bed with the patient). Plug the cable coming from the Nicolet stimulator to the lightning bolt input (if there are two breakout boxes then you need to plug one into each breakout box):



1	Base to Breakout Main Cable Connection. Labeled with
2	Cortical Stimulator Adapter Connection. Labeled with

Select **DSM stimulation** in the **Control** menu during acquisition to display the Cortical stimulation control window:

Status of cortical stimulator control interface → Stimulator ready

Operating mode → Standard mode
Stimulation through amplifier

DSM control - Manual selection of electrodes to stimulate → Electrodes: C4, C5

DSM control - Preset selection of electrodes to stimulate → C2 : C3
C3 : C4
C4 : C5
C5 : C6
Right-click here to configure presets.

Settings for Nicolet Cortical stimulator → Pulse Frequency: 60 Hz
Pulse Duration: 300 μ s
Train Duration: 0.2 s
Stimulus: 5.0 nA

Confirmation of stimulation current applied to the electrodes → Delivered Stimulation: 0.0 mA
Refer to Warning Below*

DSM operating mode and status → Automatic

Manual DSM relay control in Semi-automatic or Manual modes → Close relays

Nicolet Cortical stimulator internal stimulation test → Check Stim

Pressing this button will send a single pulse attempting to bloc seizure onset → Total Disrupt

Start stimulation → Start

Manually stop stimulation before preset train duration → Stop



* When stimulation is initiated and stopped from the controlling software, it is possible that half the stimulation amplitude will be recorded as a result of the timing of opening Switch Matrix Relays while the stimulator is still sending stimulus and measuring delivered output.